

SAP Vendor Sync API

Integration Guide for SAP Team

System: **Remittance Tool**

Module: **SAP Vendor Sync**

Version: **v1.1**

Doctype: **Remittance Vendor**

Date: **2026-06-11**

Audience: **SAP Team**

1. Overview

This API lets the SAP team push vendor lifecycle events into the Remittance Tool so the local `Remittance Vendor` master stays in sync with SAP.

What's new in v1.1 (2026-06-11)

- Full set of **SAP technical field names** (LIFNR, NAME, BEZEI, REGIO, STCD3, J_1IPANNO, VGCLASS, STR_SUPPL1...3, STREET, LOCATION, CITY1, POST_CODE1, COUNTRY, TELF1, TELF2, SMTP_ADDR, REMARK, ERDAT, BUKRS, ZCOUNT, WAERS, KTOKK, J_1KFTIND, ZTERM, TEXT1, AKONT) are now accepted directly — no need to translate to English labels before posting.
- `block_vendor` and `unblock_vendor` now also accept SAP raw field names (e.g. send `LIFNR` + `BUKRS` instead of `vendor_code` + `c_code`).
- New **SAP Mapper** doctype (Desk → SAP Mapper → Remittance Vendor) lets the dev/admin add or change field aliases at runtime — no code deploy needed. Cache auto-invalidates on save.

Endpoint	Method	Purpose
<code>upsert_vendor</code>	POST	Create new vendor or update existing (idempotent)
<code>block_vendor</code>	POST	Mark vendor as disabled (soft block)
<code>unblock_vendor</code>	POST	Clear the disabled flag
<code>upsert_vendor</code>	GET	Reachability ping (no auth)

Vendor identity is composite: `(vendor_code, c_code)`.
The same `vendor_code` can exist under different `c_code` (company codes) — these are distinct records. Both fields are required for every operation.

Base path

```
/api/method/remittance_tool.remittance_tool.api.v1.vendor.<endpoint>
```

2. Authentication setup

Step 1 — Get API Key & Secret

- Login to Frappe Desk as the SAP service user (e.g. `sap-sync@meril.local`).
- Top-right avatar → **My Settings** → scroll to **API Access**.
- Click **Generate Keys**. Copy the `api_key` and `api_secret` immediately — secret is shown only once.

Step 2 — Postman environment (recommended)

Variable	Value
<code>base_url</code>	<code>https://<your-site></code>
<code>api_key</code>	(paste from Step 1)
<code>api_secret</code>	(paste from Step 1)

Step 3 — Common Postman settings (every request)

Tab	Setting
Authorization	Type: Basic Auth · Username: <code>{{api_key}}</code> · Password: <code>{{api_secret}}</code>
Headers	<code>Content-Type: application/json</code> <code>Accept: application/json</code>
Body	Type: raw → JSON

3. Common response envelope

Every write endpoint returns the same shape. HTTP status is always `200` — inspect the body's `status` / `failed` fields.

```

{
  "message": {
    "status": "success",          // "success" | "partial" | "failed"
    "count": 2,
    "succeeded": 2,
    "failed": 0,
    "results": [
      { "status": "success", "vendor_code": "V0001", "c_code": "1000",
        "name": "V0001-1000", "action": "created", "ignored_fields": [] },
      { "status": "success", "vendor_code": "V0002", "c_code": "1000",
        "name": "V0002-1000", "action": "updated", "ignored_fields": [] }
    ]
  }
}

```

Field	Meaning
status	success = all rows ok · partial = some failed · failed = all failed
action	created · updated · blocked · unblocked
ignored_fields	Payload keys that didn't match any known vendor field (debugging aid)

Per-row commits: Every row is committed independently using a savepoint. A single bad vendor in a bulk batch fails alone — the rest of the batch still saves.

4. Test cases

TC-01 Create vendor — full SAP payload (28 fields)

POST `upsert_vendor`

`{{base_url}}/api/method/remittance_tool.remittance_tool.api.v1.vendor.upsert_vendor`

Request body

```
{
  "Vendor": "V100001",
  "Name": "Acme Imports LLC",
  "State": "Maharashtra",
  "State Code": "27",
  "GSTN No": "27AAAPL1234C1Z5",
  "PAN No": "AAAPL1234C",
  "Vendor GST Classification": "Regular",
  "Address01": "Plot 45, Andheri Industrial Estate",
  "Address02": "Off Veera Desai Road",
  "Address03": "Andheri West",
  "Address04": "Mumbai",
  "Address05": "India",
  "City": "Mumbai",
  "Pincode": "400053",
  "Country": "India",
  "Contact No": "+91-9876543210",
  "Alterenate No": "+91-2226789876",
  "Email-Id": "ap@acme.example",
  "Remark": "Onboarded via SAP sync on 2026-05-02",
  "Created On": "2026-04-15",
  "C.Code": "1000",
  "Count": 1,
  "Currency": "INR",
  "Account Group": "ZIND",
  "Type of Industr": "Manufacturing",
  "Payment Term": "NT30",
  "Payment Term Desc.": "Net 30 days",
  "Reconciliation Acco": "200110"
}
```

Expected response (HTTP 200)

```
{ "message": { "status": "success", "count": 1, "succeeded": 1, "failed": 0,
  "results": [ { "status": "success", "vendor_code": "V100001", "c_code": "1000",
    "name": "V100001-1000", "action": "created", "ignored_fields": [] } ] } }
```

Pass criteria

- `action` = `"created"`
- Record visible in Desk: **Remittance Vendor** list with name `V100001-1000`
- All 28 fields populated correctly in the form view
- `ignored_fields` is empty

TC-02 Update existing vendor — partial payload

POST `upsert_vendor` (same URL as TC-01)

Request body

```
{
  "Vendor": "V100001",
  "C.Code": "1000",
  "Email-Id": "ap-new@acme.example",
  "Contact No": "+91-9000000000",
  "Remark": "Updated email and phone via SAP sync"
}
```

Pass criteria

- `action` = `"updated"`
- Only the 3 fields sent are changed in the record
- Other fields (Name, GSTN, PAN, Address, etc.) remain as set in TC-01

TC-03 Same vendor_code, different c_code — creates a new record

POST `upsert_vendor`

Request body

```
{
  "Vendor": "V100001",
  "Name": "Acme Imports LLC - Plant 2",
  "C.Code": "2000",
  "Country": "India",
  "Currency": "INR",
  "Email-Id": "ap-plant2@acme.example"
}
```

Pass criteria

- `action` = `"created"` (NOT updated — this is a different (vendor, c_code) pair)
- New record name = `V100001-2000`
- Original record `V100001-1000` from TC-01 is untouched
- Desk list now shows 2 distinct records with the same vendor code

TC-04 Bulk create

POST `upsert_vendor`

Request body

```
{
  "vendors": [
    {
      "Vendor": "V100002", "Name": "Bravo Trading Co",
      "C.Code": "1000", "Country": "United States", "Currency": "USD" },
    {
      "Vendor": "V100003", "Name": "Charlie Exports",
      "C.Code": "1000", "Country": "Germany", "Currency": "EUR" }
  ]
}
```

Pass criteria

- `count: 2`, `succeeded: 2`, `failed: 0`
- Both records present in Desk

TC-05 Block vendor

POST `block_vendor`

```
{{base_url}}/api/method/remittance_tool.remittance_tool.api.v1.vendor.block_vendor
```

Request body — single

```
{
  "vendor_code": "V100001",
  "c_code": "1000",
  "reason": "Tax compliance hold per SAP"
}
```

Request body — bulk

```
{
  "vendors": [
    { "vendor_code": "V100001", "c_code": "1000" },
    { "vendor_code": "V100001", "c_code": "2000" }
  ],
  "reason": "Mass review hold"
}
```

Pass criteria

- `action` = `"blocked"`
- Record's **Disabled** checkbox is now in Desk
- **Block Reason** field shows the reason text
- Vendor no longer appears in Form 15CB → Vendor link dropdown

TC-06 Unblock vendor

POST `unblock_vendor`

```
{{base_url}}/api/method/remittance_tool.remittance_tool.api.v1.vendor.unblock_vendor
```

Request body

```
{
  "vendor_code": "V100001",
  "c_code": "1000"
}
```

Pass criteria

- `action` = `"unblocked"`
- **Disabled** checkbox cleared, **Block Reason** empty
- Vendor again selectable in Form 15CB

TC-07 Upsert with pure SAP raw field names

POST `upsert_vendor`

Verifies SAP can post the exact SAP table column names (no translation needed).

Request body

```
{
  "LIFNR": "0000010602",
  "NAME": "TEST STCD5",
  "BEZEI": "24-Gujarat",
  "REGIO": "06",
  "STCD3": "0",
  "J_1IPANNO": "",
  "VGCLASS": "Registered",
  "STR_SUPPL1": "",
  "STR_SUPPL2": "",
  "STREET": "VAPI",
  "STR_SUPPL3": "",
  "LOCATION": "",
  "CITY1": "VAPI",
  "POST_CODE1": "396102",
  "COUNTRY": "IN",
  "TELF1": "9899999999",
  "TELF2": "",
  "SMTP_ADDR": "STCD5@MERIL.COM",
  "REMARK": "GSTN No is Invalid",
  "ERDAT": "2025-07-14",
  "BUKRS": "7000",
  "ZCOUNT": 1,
  "WAERS": "INR",
  "KTOKK": "ZMAT",
  "J_1KFTIND": "",
  "ZTERM": "T028",
  "TEXT1": "60% within 15 days",
  "AKONT": "0013301700"
}
```

Pass criteria

- `action` = `"created"` (first run) or `"updated"` (subsequent)
- Record name = `0000010602-7000`
- `ignored_fields` is **empty** — every SAP column mapped to a Remittance Vendor field
- Open record in Desk: `state` = "24-Gujarat", `state_code` = "06", `email_id` = "STCD5@MERIL.COM", `created_on` = "2025-07-14", `c_code` = "7000"

TC-08 Block / Unblock with SAP raw keys

POST `block_vendor`, `unlock_vendor`

Verifies the lifecycle endpoints also accept `LIFNR` + `BUKRS` instead of `vendor_code` + `c_code`.

Request body — single

```
{
  "LIFNR": "0000010602",
  "BUKRS": "7000",
  "reason": "Block via SAP raw keys"
}
```

Request body — bulk

```
{
  "vendors": [
    { "LIFNR": "0000010602", "BUKRS": "7000" },
    { "LIFNR": "0000010603", "BUKRS": "7000" }
  ],
  "reason": "Mass block via SAP raw keys"
}
```

Pass criteria

- Same outcome as TC-05/TC-06 — `action` = `"blocked"` / `"unblocked"` as applicable
- No errors about "Vendor code is required" — the API resolves `LIFNR` → `vendor_code` and `BUKRS` → `c_code` automatically

TC-09 Bulk upsert — real SAP payload (multiple vendors)

POST `upsert_vendor`

End-to-end test of a realistic batch the SAP team will actually push. Mix of valid and edge rows; checks per-row success isolation.

Request body


```

{
  "vendors": [
    {
      "LIFNR": "0000010602", "NAME": "TEST STCD5",
      "BEZEI": "24-Gujarat", "REGIO": "06",
      "STCD3": "0", "VGCLASS": "Registered",
      "STREET": "VAPI", "CITY1": "VAPI",
      "POST_CODE1": "396102", "COUNTRY": "IN",
      "TELF1": "9899999999", "SMTP_ADDR": "STCD5@MERIL.COM",
      "ERDAT": "2025-07-14", "BUKRS": "7000",
      "WAERS": "INR", "KTOKK": "ZMAT",
      "ZTERM": "T028", "TEXT1": "60% within 15 days",
      "AKONT": "0013301700"
    },
    {
      "LIFNR": "0000010623", "NAME": "Testing Account Holder",
      "BEZEI": "24-Gujarat", "REGIO": "06",
      "STCD3": "22ABCDG1234D1Z2", "J_1IPANNO": "ABCDE1233D",
      "VGCLASS": "Registered", "STREET": "vapi",
      "CITY1": "Surat", "POST_CODE1": "395001",
      "COUNTRY": "IN", "TELF2": "7990088123",
      "SMTP_ADDR": "xyz@testing.com", "ERDAT": "2025-10-30",
      "BUKRS": "7000", "WAERS": "INR",
      "KTOKK": "ZMAT", "ZTERM": "Y005",
      "AKONT": "0013301300"
    },
    {
      "LIFNR": "0000010612", "NAME": "Qingdao Hafa Aile Biological Engin",
      "REGIO": "ZZ", "VGCLASS": "Not registered",
      "STR_SUPPL1": "Qingdao , China , Shandong",
      "STREET": "Building 2, No. 48 Hong Kong West Road,",
      "CITY1": "Qingdao", "POST_CODE1": "266061",
      "COUNTRY": "CN", "TELF2": "1595338654",
      "SMTP_ADDR": "gamila@hynaut.cn", "ERDAT": "2025-08-25",
      "BUKRS": "7000", "WAERS": "INR",
      "KTOKK": "ZMAT", "ZTERM": "T031",
      "AKONT": "0013301700"
    }
  ]
}

```

Pass criteria

- `count: 3`, `succeeded: 3`, `failed: 0` — assuming **both** `IN` and `CN` are present in Remittance Country master.
- If `CN` is missing from the master: `count: 3`, `succeeded: 2`, `failed: 1`. The failed row reports *"Country 'CN' does not exist..."* — the other two still save (per-row commit isolation).
- All saved records have `ignored_fields: []`.

TC-10 Mixed-alias payload (SAP raw + English mixed)

POST `upsert_vendor`

Demonstrates that the API normalises case, underscores, dots, and spaces — one payload can mix SAP raw and English labels without issue.

Request body

```
{
  // SAP raw for primary keys + tax fields
  "LIFNR": "V200001",
  "BUKRS": "1000",
  "STCD3": "27ABCDE1234F1Z5",
  "J_IIPANNO": "ABCDE1234F",
  // English labels for the rest
  "Name": "Mixed Alias Test Vendor",
  "City": "Mumbai",
  "Pincode": "400001",
  "Country": "India",
  "Email-Id": "mixed@test.example",
  "Currency": "INR",
  // Lowercase + underscore variant (also valid)
  "account_group": "ZIND"
}
```

Pass criteria

- `action: "created"`, `ignored_fields: []`
- Record `V200001-1000` shows all values correctly mapped — `pan` =ABCDE1234F, `gstn_no` =27ABCDE1234F1Z5, `email_id` =mixed@test.example, `account_group` =ZIND
- Confirms case-insensitive matching and mixing aliases in a single payload is supported.

5. Field mapping (SAP → Remittance Tool)

The API accepts SAP-style column headers and our internal snake_case names. Matching is case-insensitive; `_`, `.`, and spaces are normalized.

SAP / payload key	Stored as
LIFNR , Vendor , vendor code	vendor_code
NAME , Name , NAME1 , vendor name	vendor_name
BEZEI , State	state
REGIO , State Code	state_code
STCD3 , GSTN No , GST No	gstn_no
J_1PANN0 , PAN No , PAN	pan
VGCLASS , Vendor GST Classification	vendor_gst_classification
STR_SUPPL1 , Address01	address01
STR_SUPPL2 , Address02	address02
STREET , Address03	address03
STR_SUPPL3 , Address04	address04
LOCATION , Address05	address05
CITY1 , City , Town City District	city_district
POST_CODE1 , Pincode , Pin Code	pincode
COUNTRY , Country (name / ISO code / country code)	country (resolved against Remittance Country master)
TELF1 , Contact No	contact_no
TELF2 , Alterenate No , Alternate No	alternate_no
SMTP_ADDR , Email-Id , Email Id	email_id
REMARK , Remark	remark
ERDAT , Created On (YYYY-MM-DD, DD-MM-YYYY, DD/MM/YYYY, YYYYMMDD)	created_on
BUKRS , C.Code , C Code	c_code [required, part of unique key]
ZCOUNT , Count	count
WAERS , Currency	currency
KTOKK , Account Group	account_group
J_1KFTIND , Type of Industr , Type of Industry	type_of_industry
ZTERM , Payment Term , Terms of Payment	terms_of_payment
TEXT1 , Payment Term Desc.	payment_term_description
AKONT , Reconciliation Acco , Reconciliation Account	reconciliation_account

Runtime-configurable mappings: the dev/admin can add or change aliases without a code deploy — open **Desk → SAP Mapper → Remittance Vendor** and edit the table. The cache invalidates automatically on save, so the new alias takes effect within seconds. Use this when SAP introduces a new column or renames one.

Unknown keys are silently dropped and reported in `ignored_fields` per row. If you see your field name there, the alias is missing — raise it with the dev team, or add it directly in SAP Mapper.

6. Error reference

Error message	Meaning & fix
Authentication required (HTTP 401)	Basic Auth header missing/wrong. Re-check api_key/api_secret.
Vendor code is required	Payload missing Vendor / vendor_code .
Company Code (C.Code) is required	Payload missing C.Code / c_code . Both fields form the composite key.
Vendor name is required for new vendor X/1000	New record but no Name field.
Country '...' does not exist in Remittance Country	Country master must contain the value (by name, code, or ISO). Add to master first.
Vendor V123 (C.Code 1000) not found	Block/Unblock target doesn't exist with that key combo.
Duplicate entry '...' for key 'PRIMARY'	Schema not reloaded after autoname change. Run: bench --site <site> reload-doctype "Remittance Vendor"
ignored_fields: [...] in result	Payload key not recognized. Either typo or alias missing — check Section 5.

Server-side traces: For deeper debugging, every per-row failure logs a full traceback under Desk → **Error Log** with title **Vendor API error (...)** .

7. SAP event → endpoint recipe

SAP event / transaction	Call this endpoint
Create new vendor (XK01)	upsert_vendor
Update vendor (XK02)	upsert_vendor
Block vendor (XK05 , XK06)	block_vendor
Unblock vendor	unblock_vendor

All endpoints are idempotent — safe to retry the same payload after a failure or timeout. No duplicate records will be produced.

8. cURL examples — for terminal/scripted tests

Alternative to Postman. Replace `BASE_URL`, `API_KEY`, `API_SECRET` with your values.

8.1 Reachability ping (no auth)

```
curl -X GET \
  "$BASE_URL/api/method/remittance_tool.remittance_tool.api.v1.vendor.upsert_vendor"
```

8.2 Upsert — single vendor (SAP raw payload)

```
curl -X POST \
  "$BASE_URL/api/method/remittance_tool.remittance_tool.api.v1.vendor.upsert_vendor" \
  -u "$API_KEY:$API_SECRET" \
  -H "Content-Type: application/json" \
  -d '{
    "LIFNR": "0000010602",
    "NAME": "TEST STCD5",
    "BEZEI": "24-Gujarat",
    "REGIO": "06",
    "STCD3": "0",
    "VGCLASS": "Registered",
    "STREET": "VAPI",
    "CITY1": "VAPI",
    "POST_CODE1": "396102",
    "COUNTRY": "IN",
    "TELF1": "9899999999",
    "SMTP_ADDR": "STCD5@MERIL.COM",
    "ERDAT": "2025-07-14",
    "BUKRS": "7000",
    "WAERS": "INR",
    "KTOKK": "ZMAT",
    "ZTERM": "T028",
    "AKONT": "0013301700"
  }'
```

8.3 Upsert — bulk (from file)

Save batch as `vendors.json`, then:

```
curl -X POST \
  "$BASE_URL/api/method/remittance_tool.remittance_tool.api.v1.vendor.upsert_vendor" \
  -u "$API_KEY:$API_SECRET" \
  -H "Content-Type: application/json" \
  --data-binary @vendors.json
```

8.4 Block vendor (SAP raw keys)

```
curl -X POST \
  "$BASE_URL/api/method/remittance_tool.remittance_tool.api.v1.vendor.block_vendor" \
  -u "$API_KEY:$API_SECRET" \
  -H "Content-Type: application/json" \
  -d '{ "LIFNR": "0000010602", "BUKRS": "7000", "reason": "Tax hold" }'
```

8.5 Unblock vendor

```
curl -X POST \
  "$BASE_URL/api/method/remittance_tool.remittance_tool.api.v1.vendor.unblock_vendor" \
  -u "$API_KEY:$API_SECRET" \
  -H "Content-Type: application/json" \
  -d '{ "LIFNR": "0000010602", "BUKRS": "7000" }'
```

Auth tip: `curl -u "$API_KEY:$API_SECRET"` sends `Authorization: Basic <base64>` automatically. No need to construct the header by hand.